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PAPER_1097: Production Scaling v24 — Vectorized REST + QCalcGeom Benchmark

Abstract

We benchmark the v24 production UQFF pipeline against a sustained throughput target of 9×10^5 calc/s using six vectorized computational kernels. This extends v23 (PAPER_1091) with: precomputed S_{26} arrays, batch 99-system evaluation, REST API overhead simulation (uqff_server.js, Port 3141), and QCalcGeom geometric correction pipeline.

\$\S\$1 Vectorization Strategy

\$\S\$1.1 Precomputed S_{26} Array

$$S_{26,k} = e^{-[\text{SSq}] \cdot k/26}, \quad k = 1, \dots, 26$$

Stored once per session, reused across all kernel evaluations. Eliminates redundant $\exp()$ calls (26 per evaluation $\rightarrow$ 0).

\$\S\$1.2 Batch Mass/Radius Arrays

For the 99-system kernel:

$$M_i = 0.1 + 2i, \quad r_i = 10^9(1 + 0.3i), \quad i = 1, \dots, 99$$

Pre-allocated arrays enable vectorized inner loops.

\$\S\$2 Six Benchmark Kernels

#	Kernel	Operation	Vectorization
1	Gravity	$\sum_k GM/r^2 \cdot S_{26,k}$	Precomputed $S_{26,k}$
2	Phonon	$\Phi_0 \cdot S_{26}^2 \cdot \beta_i$	Scalar (pre-multiplied)
3	Buoyancy	$\sum_k GM/r^2 \cdot S_{26,k} \cdot \beta_i$	Precomputed
4	99-system	$\sum_{i=1}^{99} GM_i M_\odot / r_i^2$	Batch arrays
5	REST overhead	$1/(1 + 50 \mu s)$	Latency simulation
6	QCalcGeom	$\sum_k S_{26,k} \sqrt{k+1}$	Geometric correction

\$\S\$3 Methodology

Each kernel runs $n = 5000$ iterations:

$$\text{rate}_k = \frac{n}{t_{\text{elapsed},k}}$$

$$\text{average} = \frac{1}{6} \sum_{k=1}^6 \text{rate}_k$$

$$\text{meets} = (\text{average} \geq 9 \times 10^5)$$

\$\S\$4 Expected Performance vs v23

Metric	v23 (PAPER_1091)	v24 (This paper)
Kernels	4	6 (+REST, +QCalcGeom)
S_{26}	Recomputed per call	Pre-allocated array
99-system	Inline generation	Pre-allocated batch
Target	900,000 calc/s	900,000 calc/s
Expected speedup	1.0x	> 1.5x (vectorization)

\$\S\$5 REST + QCalcGeom Pipeline

$$t_{\text{total}} = t_{\text{compute}} + t_{\text{REST}} + t_{\text{QCalcGeom}}$$

The vectorized pipeline amortizes REST overhead ($\sim 50 \mu\text{s}$ per request) across batch evaluations, achieving effective throughput:

$$\text{effective} = \frac{n_{\text{batch}}}{t_{\text{compute}} + t_{\text{REST}} + t_{\text{QCalcGeom}}} \geq 9 \times 10^5$$

References

- PAPER_1091: Production Scaling v23 Benchmark
- PAPER_1088: $F_{U,Bi,i}$ Seven-Component Decomposition
- PAPER_985: ProductionKernelFUBiCompleteCalc (CP4)
- PAPER_988: RESTFUBiEndpointCalc (CP4)